

Deep Learning Fundamentals

What is Deep Learning?

Deep Learning is a subset of Machine Learning that uses artificial neural networks with multiple layers to model complex patterns in data.

Neural Network Architecture

Components:

- Input Layer: Receives data
- Hidden Layers: Process information
- Output Layer: Produces predictions
- Weights & Biases: Learnable parameters

Common Deep Learning Models

- **Convolutional Neural Networks (CNN)**: For image processing
- **Recurrent Neural Networks (RNN)**: For sequential data
- **Long Short-Term Memory (LSTM)**: For long sequences
- **Transformers**: For NLP and attention mechanisms

Activation Functions

- ReLU (Rectified Linear Unit)
- Sigmoid
- Tanh
- Softmax

Popular Frameworks

- TensorFlow
- PyTorch
- Keras